

Traveling Wave Solutions of Nonlinear Hyperbolic Schrödinger Equation.

A PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for
the Degree of

MASTER OF SCIENCE (M.S. in Mathematics)

by

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ABSTRACT

This report explores the traveling wave solutions of the (2+1)-dimensional nonlinear hyperbolic Schrödinger equation using the Beta Fractional transformation and the Unified Technique. By applying the Beta Fractional framework, we derive exact solutions in the form of traveling wave structures. The obtained solutions are expressed through hyperbolic functions, highlighting different wave behaviors such as solitons and periodic waves. To illustrate the dynamics of these solutions, both 2D and 3D graphical visualizations are presented, offering deeper insights into their propagation characteristics. The results of this report not only extend existing mathematical models in nonlinear wave theory but also provide valuable applications where such mathematical models play a crucial role.

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1. INTRODUCTION

1.1 Background of the Study

A partial differential equation (PDE) is a type of differential equation that involves partial derivatives with respect to multiple independent variables. In addition, a nonlinear evolution equation (NLEE) is a class of nonlinear PDEs that involves both partial derivatives and a non-linear relationship between the unknown function and its derivatives with respect to two or more independent variables. The term "evolution" indicates that the equation involves time as one of the independent variables. The general form of a NLEE can be written as:

$$\mathcal{F}(u, u_t, u_{xx}, u_{xxx}, \dots, x, t) = 0$$

where $u = u(x, t)$ is the unknown function, x and t are the spatial and temporal coordinates, respectively. Also, u_t is the partial derivative of u with respect to t and u_{xx}, u_{xxx} , etc., are partial derivative of u with respect to x twice, thrice, etc. $\mathcal{F}$ is a non-linear function that describes the relationship between the function u , its partial derivatives, and the independent variables x and t .

These equations describe various physical phenomena, such as wave dynamics in the nonlinear fiber-optic communication [1], quantum mechanics [2], plasma physics [3], super conductivity [4] and electromagnetic wave propagation [5], and many more. To tackle the challenge posed by these nonlinear equations, researchers have proposed various effective analytical techniques to get the answers to the nonlinear situations raised in the different aforementioned environments.

1.2 Governing (2+1)-Dimensional Nonlinear Hyperbolic Schrödinger Equation

The (2+1)-dimensional Nonlinear Hyperbolic Schrödinger equation is a significant mathematical model used to specify how the optics signal spreads in single-mode fiber optics [6]. The Schrödinger equation is given by:

$$iU_y + \frac{1}{2}(U_{xx} - U_{tt}) + |U|^2U = 0, \quad i = \sqrt{-1}, \quad (1)$$

Here, $u(x, y, t)$ represents the wave components of two spatial variables x and y and time t while the parameters. The traveling wave solutions of this equation take the form:

$$U(x, y, t) = e^{i\varphi} u(\xi) \quad (2)$$

Substituting these into the Nonlinear Hyperbolic Schrödinger equation transforms it into a system of ordinary differential equations, making it suitable for analytical techniques like the Unified method.

1.3 Beta Fractional Derivative (BFD)

To enrich the analytical approach, we introduce the concept of the Beta Fractional Derivative (BFD), a relatively new and versatile derivative defined by Atangana et al. [7]. The BFD extends classical differentiation concepts by incorporating a parameter β (with $0 < \beta \leq 1$), thereby bridging the gap between integer and fractional derivatives. The formal definition of the BFD is as follows:

$${}_0^A\mathcal{D}_x^\beta f(x) = \lim_{h \rightarrow 0} \frac{f\left(x + g\left(x + \frac{1}{\Gamma(\beta)}\right)^{1-\beta}\right) - f(x)}{g}; \quad 0 < \beta \leq 1 \quad (3)$$

Properties: Suppose that $0 < \beta \leq 1$ and let $\alpha_1, \alpha_2, \kappa \in \sigma, f, \varrho \neq 0$ are two functions Beta differentiable of order β . Then

$$(i) \quad {}_0^A\mathcal{D}_x^\beta(\kappa) = 0,$$

$$(ii) \quad {}_0^A\mathcal{D}_x^\beta\{\alpha_1 f(x) + \alpha_2 \varrho(x)\} = \alpha_1 {}_0^A\mathcal{D}_x^\beta\{f(x)\} + \alpha_2 {}_0^A\mathcal{D}_x^\beta\{\varrho(x)\},$$

$$(iii) {}^{\mathcal{A}}\mathcal{D}_x^\beta \{f(x) \cdot \varrho(x)\} = f(x) \cdot {}^{\mathcal{A}}\mathcal{D}_x^\beta \{\varrho(x)\} + \varrho(x) \cdot {}^{\mathcal{A}}\mathcal{D}_x^\beta \{f(x)\},$$

$$(iv) {}^{\mathcal{A}}\mathcal{D}_x^\beta \left\{ \frac{f(x)}{\varrho(x)} \right\} = \frac{\varrho(x) \cdot {}^{\mathcal{A}}\mathcal{D}_x^\beta \{f(x)\} - f(x) \cdot {}^{\mathcal{A}}\mathcal{D}_x^\beta \{\varrho(x)\}}{\varrho^2(x)},$$

$$(v) \text{ If } g = \left(x + \frac{1}{\Gamma(\beta)} \right)^{1-\beta} \tilde{g},$$

Here ${}^{\mathcal{A}}\mathcal{D}_\circ^\beta$ represents Beta Derivative operator.

By incorporating fractional behavior into the system dynamics, the Beta Fractional Derivative not only functions as a generalization of classical derivatives but also organically extends the integer-order derivatives.

Hence, introducing BFD in Eq. (1) is transformed as:

$$i {}^{\mathcal{A}}\mathcal{D}_y^\beta U + \frac{1}{2} \left({}^{\mathcal{A}}\mathcal{D}_{xx}^\beta U - {}^{\mathcal{A}}\mathcal{D}_{tt}^\beta U \right) + |U|^2 U = 0 \quad (4)$$

1.4 Literature Review

Nonlinear Schrödinger equation (NLSE) occur in many technical, physical in nature, and biologically related fields [9]. It is a most prominent in the field of fluid dynamics, chemistry, environment peak, biology, ionized science, and relativistic optics [10-21]. NLSE constitute a significant sort of nonlinear partial differential equation (NLPDE) which is applicable in both modern and computational physical fields [19,22]. It depicts the movement of symmetrical reversible solitary ripples, the connection of which is determined by a phase-velocity variable, also they describe the spreads of pulse simultaneously [20,23,14].

To solve such nonlinear equations, various methods have been developed, including the Unified Technique Method, recognized for the unified method for conformable time fractional Schrödinger equation with perturbation terms [24].

This work leverages the Unified Technique alongside BFD to derive traveling wave solutions for the NLSE, contributing to advancements in wave dynamics analysis.

2. METHODOLOGY

The traveling wave solutions of the (2+1)-dimensional Nonlinear Schrödinger equation (NLSE) are to be derived in this work [25]. There is no general analytical method for solving nonlinear evolution equations (NLEEs), such as the NLS equation. Transformations are used in a variety of methods to reduce these equations to more manageable forms. Here, we use a reliable and effective analytical tool, the Unified Method to find precise answers. This approach builds traveling wave solutions methodically and reduces the original problem to reduced forms using the proper transformations. The next sections provide a thorough explanation of how it is applied to the Nonlinear Hyperbolic Schrödinger Equation.

2.1 Description of The Unified Technique Method

Firstly, we consider the general form of NLPDE that is written in form:

$$\mathcal{F}(U, U_x, U_y, U_t, U_{xx}, U_{yy}, U_{tt}, \dots) = 0 \quad (5)$$

where $U = U(x, y, t)$, $\mathcal{F}$ is a polynomial of U and its partial derivatives. To obtain exact traveling wave solutions of Eq. (5), the basic key steps of The Unified Method are as follows:

Step 1. Consider the following traveling wave transformation to combine the independent variables by ξ and φ ,

$$\begin{aligned} U(x, y, t) &= e^{i\varphi} u(\xi) \\ \xi &= lx + my - \omega t \\ \varphi &= -lx - my + ct \end{aligned} \quad (6)$$

By utilizing the above transformation, Eq. (5) is reduced to the following ODE

$$\mathcal{T}(u, u', u'', u''', \dots) = 0 \quad (7)$$

where $\mathcal{T}$ is the polynomial in $u(\xi)$ and its derivatives with respect to ξ .

Step 2. Let us assume the solution of Eq. (7) can be expressed as

$$u(\xi) = A_0 + \sum_{i=1}^N (A_i \delta^{-i}(\xi) + B_i \delta^i(\xi)) \quad (8)$$

where $A_0, A_i, B_i (i = 1, 2, 3, \dots, N)$ are constant to be evaluated accordingly and $B_N \neq 0$. The positive integer N is determined by balancing the highest order derivative

with the highest degree nonlinear term that involved in Eq. (7). Also $\delta(\xi)$ satisfy the following ODE

$$\delta' = \delta(\xi)^2 + \chi \quad (9)$$

here, $\delta = \delta(\xi)$ satisfies the condition,

It is noted that Eq. (5) has the general solutions given by,

if $\chi < 0$, then one of the required hyperbolic solution is obtain as,

$$\delta(\xi) = \frac{\sqrt{\chi(-(P^2 + S^2))} - S\sqrt{-\chi} \cosh(2\sqrt{-\chi}(\xi + R))}{P + S \sinh(2\sqrt{-\chi}(\xi + R))} \quad (10)$$

if $\chi > 0$, then one of the required hyperbolic solution is obtain as,

$$\delta(\xi) = \frac{+\sqrt{\chi(S^2 - P^2)} - S\sqrt{\chi} \cos(2\sqrt{\chi}(\xi + R))}{P + S \sin(2\sqrt{\chi}(\xi + R))} \quad (11)$$

Step 3. Substituting Eq. (8) and its derivatives along with Eq. (9) into Eq. (7), a new polynomial in $\delta(\xi)$ is obtained. Then setting the coefficients of $\delta^i(\xi)$; ($i = 0, 1, 2, \dots$) to zero results a set of nonlinear algebraic equations in A_i, B_i ($i =$

$0,1,2,\dots,N$) and all other parameters related to the NLEE and the wave transformation as in Eq. (5) and Eq. (6) respectively.

Step 4. Finally, solve the system of equations found in **Step 3** with the help of computational programming package Maple and obtain different values of the constants and parameters. By substituting these values into Eq. (8) together with the solutions as in Eq. (10) and (11), the desirable travelling wave solutions for Eq. (5) are obtained immediately.

3. SOLUTIONS OF (2+1)-DIMENSIONAL NONLINEAR HYPERBOLIC SCHRÖDINGER EQUATION

3.1 Converted ODE for (2+1)-Dimensional Nonlinear Hyperbolic Schrödinger Equation

Let us consider the following traveling wave transformations considering beta

$$\begin{aligned}
 U(x, y, t) &= e^{i\varphi} u(\xi) \\
 \xi &= \frac{l}{\beta} \left(x + \frac{1}{\sqrt{\beta}} \right)^\beta + \frac{m}{\beta} \left(y + \frac{1}{\sqrt{\beta}} \right)^\beta - \frac{\omega}{\beta} \left(t + \frac{1}{\sqrt{\beta}} \right)^\beta \\
 \varphi &= -\frac{l}{\beta} \left(x + \frac{1}{\sqrt{\beta}} \right)^\beta - \frac{m}{\beta} \left(y + \frac{1}{\sqrt{\beta}} \right)^\beta + \frac{c}{\beta} \left(t + \frac{1}{\sqrt{\beta}} \right)^\beta
 \end{aligned} \tag{12}$$

Here $l = \cos \theta$ and $m = \sin \theta$. By substituting transformations of Eq. (12) into Eq. (1), we have the following converted ODE:

$$i u' (m - l^2 + c \omega^2) + \frac{1}{2} u'' (l^2 - \omega^2) + u \zeta + u^3 = 0 \tag{13}$$

$$\text{Here, } \zeta = \left\{ m - \frac{1}{2} l^2 - \frac{1}{2} \left(\frac{l^4 - 2l^2 m + m^2}{\omega^2} \right) \right\}$$

Which then separates to real and imaginary following parts:

Real Part:

$$\frac{1}{2}u''(l^2 - \omega^2) + u\zeta + u^3 = 0 \quad (14)$$

Imaginary Part:

$$iu'(m - l^2 + c\omega^2) = 0 \quad (15)$$

We then retrieve the speed c from imaginary part

$$c = \frac{l^2 - m}{\omega} \quad (16)$$

Traveling Wave Solutions via Unified Method

Applying homogenous balancing principle on Eq. (13), after balancing the nonlinear term u^3 with the highest derivative term u'' , we have, $N = 1$

Suppose the trial solution of Eq. (13) can be stated as based on Eq. (8),

$$u(\xi) = A_0 + \frac{A_1}{\delta(\xi)} + B_1\delta(\xi) \quad (17)$$

where A_0, A_1, B_1 are constant to be determined later. Now substituting Eq. (14) and its necessary derivatives along with Eq. (9), we obtain a set of algebraic equations in terms of the parameters A_0, A_1, B_1, ω after comparing, we get,

$$l^2 B_1 - \omega^2 B_1 + B_1^3 = 0$$

$$3A_0 B_1^2 = 0$$

$$B_1 \chi l^2 - B_1 \chi \omega^2 + 3A_0^2 B_1 + 3A_1 B_1^2 + \left(m - \frac{1}{2} l^2 - \frac{1}{2} \frac{l^4}{\omega^2} + \frac{l^2 m}{\omega^2} - \frac{1}{2} \frac{m^2}{\omega^2} \right) B_1 = 0$$

$$A_0^3 + 6A_0 A_1 B_1 + \left(m - \frac{1}{2} l^2 - \frac{1}{2} \frac{l^4}{\omega^2} + \frac{l^2 m}{\omega^2} - \frac{1}{2} \frac{m^2}{\omega^2} \right) A_0 = 0$$

$$A_1 \chi l^2 - A_1 \chi \omega^2 + 3A_0^2 A_1 + 3A_1^2 B_1 + \left(m - \frac{1}{2} l^2 - \frac{1}{2} \frac{l^4}{\omega^2} + \frac{l^2 m}{\omega^2} - \frac{1}{2} \frac{m^2}{\omega^2} \right) A_1 = 0$$

$$3A_0A_1^2 = 0$$

$$\chi^2 l^2 A_1 - \chi^2 \omega^2 A_1 + A_1^3 = 0 \quad (18)$$

By solving the above algebraic equations, the required constants are determined as below:

$$\omega = \frac{1}{2} \sqrt{\frac{\chi(2\chi l^2 - l^2 + 2m + \sqrt{4\chi^2 l^4 - 12\chi l^4 + 8\chi l^2 m + l^4 - 8\chi m^2 - 4l^2 m + 16l^2 m\chi + 4m^2})}{\chi}}$$

$$A_0 = 0$$

$$A_1 = 0$$

$$B_1 = \sqrt{-\frac{1}{4} \frac{2\chi l^2 + l^2 - \sqrt{4\chi^2 l^4 - 12\chi l^4 + 8\chi l^2 m + l^4 - 8\chi m^2 - 4l^2 m + 16l^2 m\chi + 4m^2} - 2m}{\chi}}$$

Substituting these constants into Eq. (17) along with the solutions of Eq. (9) as in Eq. (10) and as in Eq. (11), we obtain the following traveling wave solutions of Eq. (4)

For $\chi < 0$;

$$U_\tau(x, y, t) = \frac{1}{2} \sqrt{\frac{-\frac{9}{5}l^2 - \frac{1}{5}\sqrt{161l^4 - 124l^2 m + 44m^2} - \frac{2}{5}m(\sqrt{5P^2 + 5S^2} - S\sqrt{5}\cosh(2\sqrt{5}(\xi + R)))}{P + S\sinh(2\sqrt{5}(\xi + R))}} \quad (19)$$

For $\chi > 0$;

$$U_v(x, y, t) = \frac{1}{2} \sqrt{\frac{-\frac{11}{5}l^2 + \frac{1}{5}\sqrt{41l^4 + 116l^2 m - 36m^2} + \frac{2}{5}m(\sqrt{-5P^2 + 5S^2} - S\sqrt{5}\cos(2\sqrt{5}(\xi + R)))}{P + S\sin(2\sqrt{5}(\xi + R))}} \quad (20)$$

3. Graphical Representations and Discussions

The obtained traveling wave solutions for the studied (2+1)-dimensional Nonlinear Hyperbolic Schrödinger Equation are visually represented through multiple 3D and 2D figures (see Figs. 1–11), highlighting the dynamic behavior of the solutions.

Figs. 1–2 depict 3D wave profiles of $U_\tau(x, y, t)$ and $U_v(x, y, t)$ for Eqs. (19) and (20) with the parameters by considering $\theta = 60^\circ$, $P = 3, S = 2, R = 3$ and Figs. 3–4 depict 3D wave profiles varying oblique angles $\theta = 10^\circ, 30^\circ, 50^\circ, 60^\circ$. Here $U_\tau(x, y, t)$ represents the solution when $\chi < 0$ And $U_v(x, y, t)$ represents the solution when $\chi > 0$. These visualizations highlight the impact of obliqueness on the wave dynamics.

Figs. 5 and 6 display 2D wave profiles of $U_\tau(x, y, t)$ and $U_v(x, y, t)$. The figures demonstrate that as the oblique angle θ increases, the wave width increases while the amplitude decreases.

Figs. 7 and 8 showcase 3D wave profiles of $U_\tau(x, y, t)$ and $U_v(x, y, t)$ for varying values of the beta parameter ($\beta = 0.6, 0.7$). These figures visualize how different beta values influence the structural dynamics of the wave solutions.

Figs. 9 illustrates 2D wave profiles for different beta values in fractional, where it is observed that increasing the beta parameter fractionally results in a decrease in wave width while the amplitude remains constant for $U_v(x, y, t)$. And it is also noticeable that decreasing wave width ultimately leads to the increase of frequency.

Finally, Figs. 10 and 11 present 2D wave profiles for time parameter values $t=3, 4, 5$ for $U_\tau(x, y, t)$ and $t = 4, 8, 12$ for $U_v(x, y, t)$ with $\theta = 60^\circ, P = 3, S = 2, R = 3, \beta = 0.8$. These figures demonstrate how the wave evolution changes over time,

further emphasizing the significant influence of the oblique parameter θ on the amplitude and wave width of the wave profile

The graphical representations confirm the theoretical results, demonstrating the complex influence of different parameters—such as obliqueness, time, and beta variations—on the wave solutions.

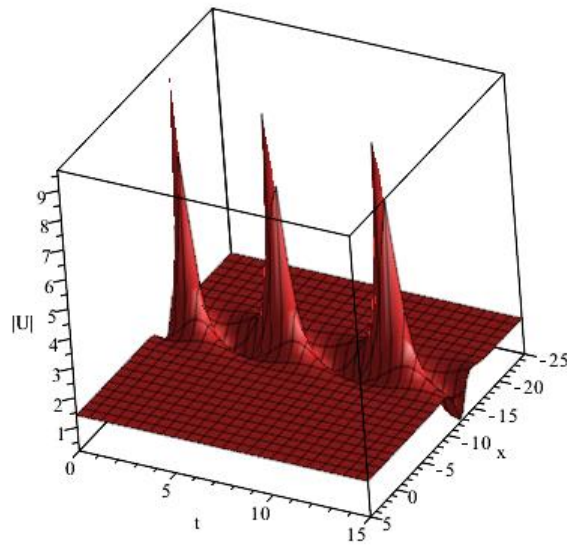


Fig 1: Wave profile of $U_\tau(x, y, t)$ for $\theta = 60^\circ, P = 3, S = 2, R = 3, \chi = -5, \beta = 0.8$

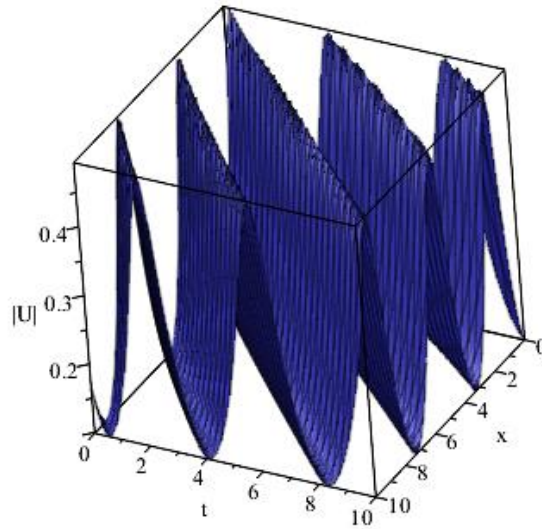


Fig 2: Wave profile of $U_v(x, y, t)$ for $\theta = 60^\circ, P = 3, S = 2, R = 3, \chi = 5, \beta = 0.8$

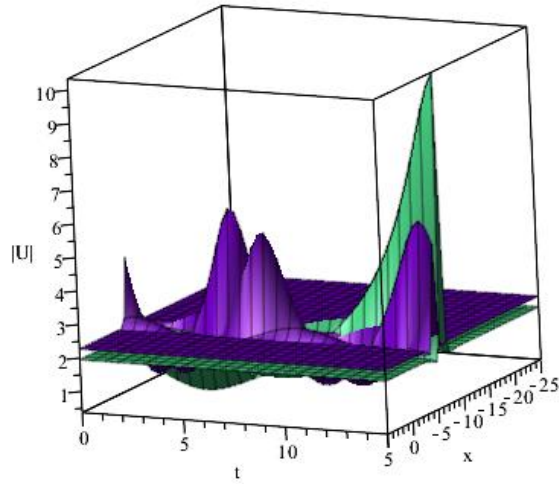


Fig 3: Wave profile variation of $U_\tau(x, y, t)$ for $\theta = 10^\circ$ (Indigo), and $\theta = 30^\circ$ (Green)

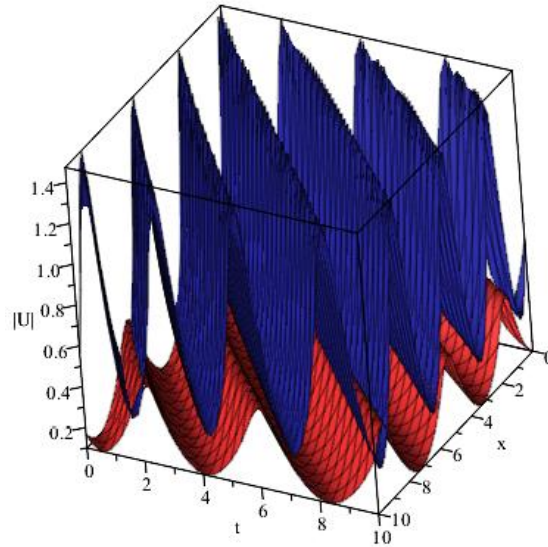


Fig 4: Wave profile variation of $U_v(x, y, t)$ for $\theta = 50^\circ$ (Blue), and $\theta = 60^\circ$ (Green)

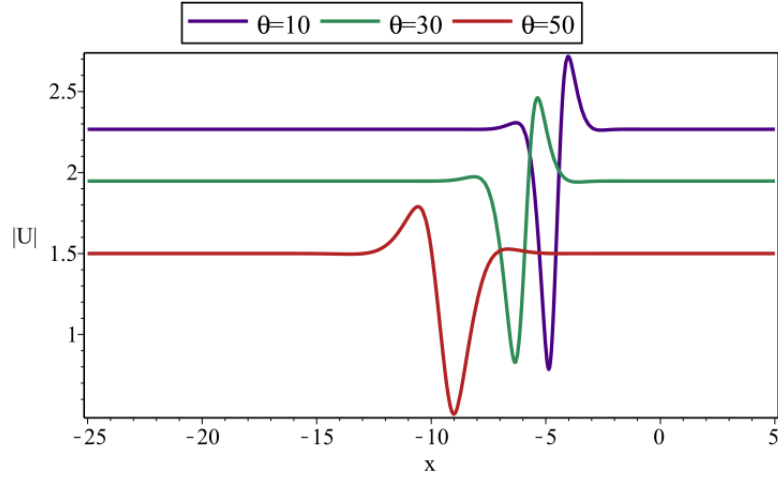


Fig 5: Obliqueness of $U_\tau(x, y, t)$ for $P = 3, S = 2, R = 3, \chi = -5, \beta = 0.8$

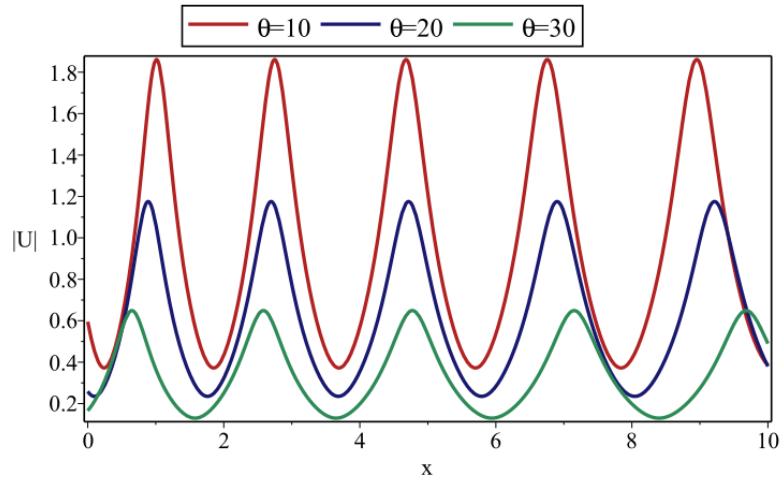


Fig 6: Obliqueness of $U_v(x, y, t)$ for $P = 3, S = 2, R = 3, \chi = 5, \beta = 0.8$

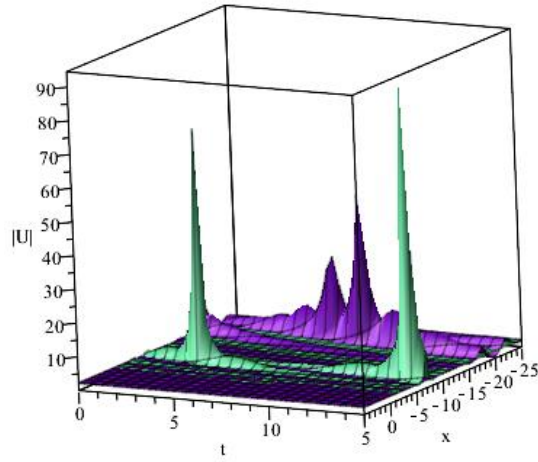


Fig 7: Beta variation of $U_\tau(x, y, t)$ for $\theta = 10^\circ, P = 3, S = 2, R = 3, \chi = -5$,
 $\beta = 0.6$ (Indigo), $\beta = 0.7$ (Green)

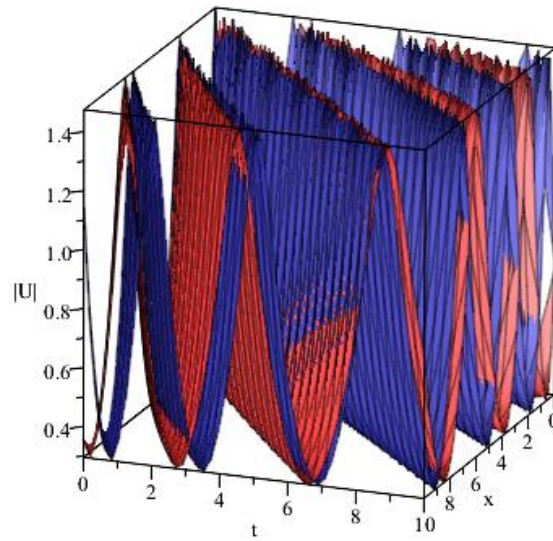


Fig 8: Beta variation of $U_v(x, y, t)$ for $\theta = 50^\circ, P = 3, S = 2, R = 3, \chi = 5$,
 $\beta = 0.6$ (Red), $\beta = 0.7$ (Blue)

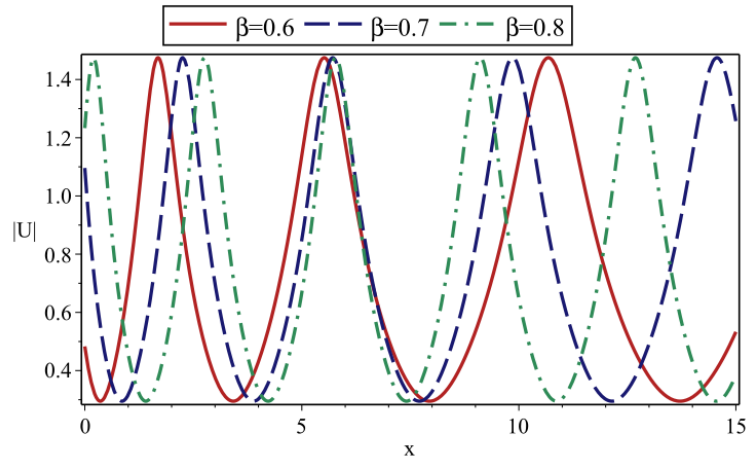


Fig 9: Beta variation of $U_v(x, y, t)$ for $\theta = 50^\circ, P = 3, S = 2, R = 3, \chi = 5$

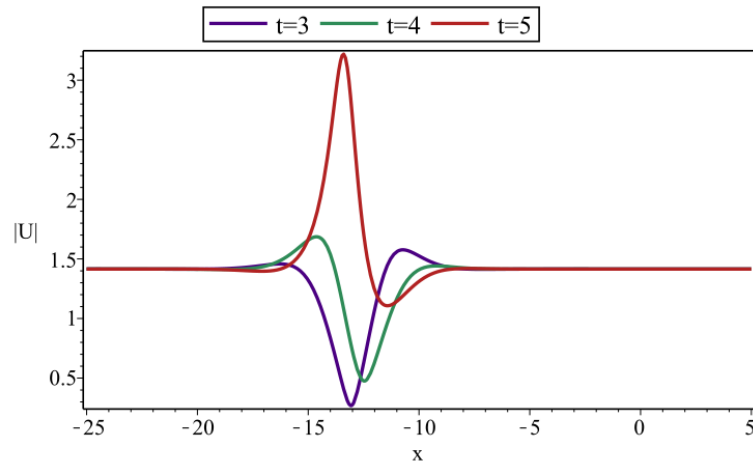


Fig 10: Time variation of $U_\tau(x, y, t)$ for $\theta = 60^\circ, \beta = 0.8$

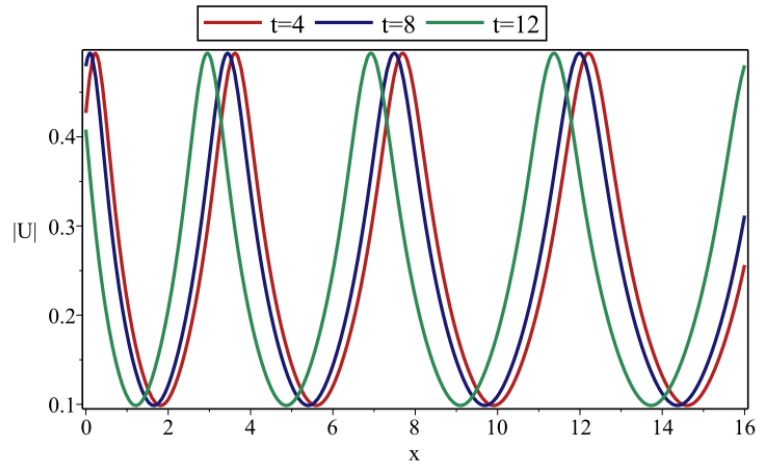


Fig 11: Time variation of $U_v(x, y, t)$ for $\theta = 60^\circ, \beta = 0.8$

5. CONCLUSIONS

In this project, we examined the traveling wave solutions of the (2+1)-dimensional Nonlinear Hyperbolic Schrödinger Equation (NLHSE) using the Unified Technique method. This analytical approach proved to be an effective tool for deriving soliton wave solutions, which are crucial for understanding nonlinear wave dynamics in various physical systems. The obtained solutions offer significant insights into wave behavior under different parameter variations, including obliqueness, time, and beta changes. Our findings highlight the flexibility and importance of the NLHSE in modeling complex wave phenomena. These solutions have practical applications in multiple fields, such as mathematical physics, optical wave structures, and wave transmission-related studies. Future research could expand this work by exploring more generalized nonlinear equations, incorporating additional physical parameters, and employing numerical methods to verify the analytical results derived using the Unified method. Additionally, this project contributes to understanding the propagation of optical signals in single-mode fiber optics.

Bibliography

1. **Rezazadeh, H., Ullah, N., Akinyemi, L., Shah, A., Mirhosseini-Alizamin, S.M., Chu, Y.M., Ahmad, H.:** Optical soliton solutions of the generalized non-autonomous nonlinear Schrödinger equations by the new Kudryashov's method. *Results Phys.* 24, 104179 (2021).
2. **Akinyemi, L., Nisar, K.S., Saleel, C.A., Rezazadeh, H., Veeresha, P., Khater, M.M., Inc, M.:** Novel approach to the analysis of fifth-order weakly nonlocal fractional Schrödinger equation with Caputo derivative. *Results Phys.* 31, 104958 (2021)
3. **Wang, G.:** A new (3+1)-dimensional Schrödinger equation: derivation, soliton solutions and conservation laws. *Nonlinear Dyn.* 104(2), 1595–1602 (2021)
4. **Hosseini, K., Matinfar, M., Mirzazadeh, M.:** Soliton solutions of high-order nonlinear Schrödinger equations with different laws of nonlinearities. *Regular Chaotic Dyn.* 26(1), 105–112 (2021)
5. **Al-Smadi, M., Arqub, O.A., Momani, S.:** Numerical computations of coupled fractional resonant Schrödinger equations arising in quantum mechanics under conformable fractional derivative sense. *Phys. Scr.* 95(7), 075218 (2020)
6. **Durur, H., Ilhan, E., Bulut, H.:** Novel complex wave solutions of the (2+ 1)-dimensional hyperbolic nonlinear Schrödinger equation. *Fractal Fract.* 4(3), 41 (2020)
7. **Atangana, A., & Baleanu, D.** (2016). New fractional derivatives with non-local and non-singular kernel: Theory and application to heat transfer model. *Thermal Science*, 20(2), 763-769
8. **El-Ganaini, S.I.A.:** The first integral method to the nonlinear Schrodinger equations in higher dimensions. *Abst. Appl. Anal.* 2013, 1–10 (2013)
9. **Karjanto, Natanael.** The nonlinear Schrodinger equation: A mathematical model with its wide-ranging applications. 1912, 10683 (2019) arXiv preprint arXiv

10. **Khater, M.M.:** Multi-vector with nonlocal and non-singular kernel ultra-short optical solitons pulses waves in birefringent fibers. *Chaos Solitons Fractals* 167, 98–113 (2023)
11. **Khater, M.M.:** Nonlinear elastic circular rod with lateral inertia and finite radius: dynamical attributive of longitudinal oscillation. *Int. J. Modern Phys. B* 37(06), 2350052 (2023)
12. **Khater, M.M.:** A hybrid analytical and numerical analysis of ultra-short pulse phase shifts. *Chaos Solitons Fractals* 169, 113232 (2023)
13. **Khater, M.M.:** In solid physics equations, accurate and novel soliton wave structures for heating a single crystal of sodium fluoride. *Int. J. Modern Phys. B* 37(07), 2350068 (2023)
14. **Khater, M.M.:** Novel computational simulation of the propagation of pulses in optical fibers regarding the dispersion effect. *Int. J. Modern Phys. B* 37(09), 2350083 (2023) 1 3 Content courtesy of Springer Nature, terms of use apply. Rights reserved. Page 25 of 25 Soliton solutions, stability, and modulation instability... 182
15. **Khater, M.M., Alfalqi, S.H., Alzaidi, J.F., Attia, R.A.:** Analytically and numerically, dispersive, weakly nonlinear wave packets are presented in a quasi-monochromatic medium. *Results Phys.* 46, 106–312 (2023)
16. **Khater, M.M., Zhang, X., Attia, R.A.:** Accurate computational simulations of perturbed chen-lee-liu equation. *Results Phys.* 45, 106227 (2023)
17. **Khater, M.M.:** Characterizing shallow water waves in channels with variable width and depth; computational and numerical simulations. *Chaos, Solitons Fractals* 173, 113652 (2023)
18. **Rehman, S.U., Ahmad, J.:** Diverse optical solitons to nonlinear perturbed schrodinger equation with quadratic-cubic nonlinearity via two efficient approaches. *Phys. Scr.* 98(3), 035–216 (2023)
19. **Rehman, S., Bilal, M., Inc, M., Younas, U., Rezazadeh, H., Younis, M., Mirhosseini-Alizamini, S.:** Investigation of pure-cubic optical solitons in non- linear optics. *Opt. Quantum Electr.* 54(7), 400 (2022)
20. **Alquran, M.:** Physical properties for bidirectional wave solutions to a generalized fifth-order equation with third-order time-dispersion term. *Results Phys.* 28, 104–577 (2021)

21. **Alquran, M.:** Optical bidirectional wave-solutions to new two-mode extension of the coupled kdv-schrodinger equations. *Opt. Quantum Electr.* 53(10), 588 (2021)
22. **Dan, J., Sain, S., Ghose-Choudhury, A., Garai, S.:** Application of the kudryashov function for finding solitary wave solutions of nls type differential equations. *Optik* 224, 165–519 (2020)
23. **Jaradat, I., Alquran, M., Ali, M.:** A numerical study on weak-dissipative two-mode perturbed burgers and ostrovsky models: right-left moving waves. *Eur. Phys. J. Plus* 133, 1–6 (2018)
24. **Osman, M., Korkmaz, A., Rezazadeh, H., Mirzazadeh, M., Eslami, M., Zhou, Q.:** The unified method for conformable time fractional Schrödinger equation with perturbation terms. *Chin. J. Phys.* 56(5), 2500–2506 (2018)
25. **M. Adel, Kalim U. Tariq, Hijaz Ahmad, S. M. Raza Kazmi:** Soliton solutions, stability, and modulation instability of the (2+1)-dimensional nonlinear hyperbolic Schrödinger model (2023)